



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

PL/I PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Programming Languages

TOTAL: 10 QUESTIONS

Q1 What are the primary design goals of PL/I?

Threat Model

Q6 How does PL/I implement object-oriented or functional programming?

Observability

Q2 How does PL/I handle type checking: static or dynamic?

Architecture

Q7 How does PL/I handle errors?

Lifecycle

Q3 How does PL/I manage memory?

Configuration

Q8 What testing tools are commonly used in PL/I?

Compliance

Q4 What is PL/I's concurrency model?

Hardening

Q9 What makes PL/I well-suited for its target domain?

Testing

Q5 How are packages and dependencies managed in PL/I?

SSO / IdP

Q10 What are common performance pitfalls in PL/I?

Tuning



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 PL/I was designed with specific priorities around

Mitigation

PL/I was designed with specific priorities around performance, safety, or developer productivity depending on its domain. These goals shape its syntax, type system, and standard library choices.

A6 PL/I supports classes and inheritance for OOP,

Observability

PL/I supports classes and inheritance for OOP, or first-class functions and immutability for FP, or both. Many modern PL/I programs blend both paradigms depending on the problem.

A2 PL/I uses its type system to catch

Primitives

PL/I uses its type system to catch errors at compile time (static) or at runtime (dynamic). This affects refactoring safety, tooling support, and the kinds of bugs that reach production.

A7 PL/I surfaces errors via exceptions, Result/Either types,

Automation

PL/I surfaces errors via exceptions, Result/Either types, or error return values. The choice impacts whether errors are checked at compile time and how calling code is structured.

A3 PL/I uses one of three approaches: manual

Hardening

PL/I uses one of three approaches: manual allocation/deallocation, a garbage collector that periodically reclaims unreachable objects, or automatic reference counting. Each has different latency and throughput tradeoffs.

A8 PL/I has a standard or widely adopted

Governance

PL/I has a standard or widely adopted test runner and assertion library. Tests are typically organized into unit, integration, and end-to-end layers with coverage reporting built in or via plugins.

A4 PL/I supports concurrency through threads, coroutines,

Gotchas

PL/I supports concurrency through threads, coroutines, async/await, or an actor model. The choice determines how I/O-bound and CPU-bound workloads are scaled and how shared state is safely accessed.

A9 PL/I excels in its domain due to

Assessment

PL/I excels in its domain due to a combination of performance characteristics, ecosystem maturity, language ergonomics, and community support that makes common tasks simple.

A5 PL/I uses a package manager and a

Federation

PL/I uses a package manager and a manifest file to declare dependencies and version constraints. A lock file pins exact versions to ensure reproducible builds across environments.

A10 Typical performance issues in PL/I include excessive

Optimization

Typical performance issues in PL/I include excessive allocations, blocking the main thread or event loop, $O(n^2)$ data structures, and missing caching. Profiling and benchmarking tools are available to diagnose them.